

Customer Churn Prediction Model and Streamlit App

Complete Machine Learning Project Report with GitHub Repository and Streamlit Cloud Deployment

Project Name	Customer Churn Predictor
Student / Developer	Jaydip Nakrani
GitHub Repository	https://github.com/Jaydipnakrani/Customer-Churn-Predictor
Live Demo	https://customer-churn-predictor-gw7ugtznrde4xssr5cwq9h.streamlit.app/
Technologies	Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit, Pickle, Git, GitHub, Streamlit Cloud

1. Project Overview

This project focuses on building a machine learning model to predict customer churn and converting the trained model into a simple web application using Streamlit. Customer churn prediction helps businesses identify customers who may stop using a service, allowing them to take early action to improve retention.

The project covers the complete practical workflow: dataset preparation, exploratory data analysis, preprocessing, model training, evaluation, model saving, Streamlit user interface development, GitHub version control, and live deployment on Streamlit Cloud.

2. Objective

- Build a small machine learning model for binary customer churn prediction.
- Perform exploratory data analysis to understand customer behavior and churn patterns.
- Preprocess data using encoding, scaling, and train-test split.
- Train and compare Logistic Regression, Random Forest, and XGBoost models.
- Convert the trained model into a Streamlit web application with user input fields.
- Display churn prediction, probability score, and a short explanation.
- Deploy the application publicly using Streamlit Cloud.

3. Dataset Description

A public Telco Customer Churn dataset was used. The dataset contains customer information such as tenure, monthly charges, total charges, contract type, internet service, payment method, and churn status. The target variable is **Churn**, which indicates whether the customer left the company or not.

4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand data quality and identify useful patterns. The dataset was checked for missing values, duplicate records, feature distributions, target distribution, and relationships between numerical and categorical variables.

- Checked null and duplicate values.
- Analyzed numerical features such as tenure, monthly charges, and total charges.
- Analyzed categorical features such as contract type, internet service, and payment method.
- Studied churn distribution to understand class balance.
- Reviewed correlations between numerical variables.

5. Data Preprocessing

The data was prepared for machine learning by converting categorical data into numerical format, scaling numerical features, and splitting the dataset into training and testing sets. The final training feature structure was saved to support correct input alignment during deployment.

- Handled missing or invalid values.
- Encoded categorical columns into numerical machine-readable columns.
- Scaled features using StandardScaler.
- Split the dataset into train and test sets.
- Saved model.pkl, scaler.pkl, and columns.pkl for deployment.

6. Model Training and Evaluation

Three classification models were trained and compared: Logistic Regression, Random Forest, and XGBoost. These models were evaluated using accuracy, precision, recall, and confusion matrix. These metrics helped compare overall correctness, churn detection quality, and false prediction behavior.

Model	Purpose
Logistic Regression	Baseline binary classification model; simple and interpretable.
Random Forest	Tree-based model that handles non-linear patterns in structured customer data.
XGBoost	Advanced boosting model often strong for tabular classification problems.

7. Best Model Summary

After model comparison, the best-performing model was selected based on the evaluation results and saved as a Pickle file for use in the Streamlit app. Tree-based models such as Random Forest and XGBoost are useful for this project because they can capture more complex customer behavior patterns than a simple linear model.

8. Streamlit Web Application

The trained churn prediction model was converted into an interactive Streamlit web application. The app allows a user to enter customer details and receive a prediction about whether the customer is likely to churn.

App Features

- User input fields for tenure, monthly charges, total charges, contract type, internet service, and payment method.
- Prediction output: Will customer churn? Yes or No.
- Churn probability displayed as a percentage.
- Short explanation of the prediction result.
- Clean and simple Streamlit user interface.
- Model, scaler, and feature columns loaded from saved .pkl files.

9. Prediction Workflow in the App

Step	Description
1	Load the trained ML model from models/model.pkl.
2	Load the scaler from models/scaler.pkl.
3	Load original training columns from models/columns.pkl.
4	Accept user input from the Streamlit interface.
5	Convert input into the same column format used during training.
6	Scale the input data using the saved scaler.
7	Predict churn result and churn probability.
8	Display prediction result with a short explanation.

10. Example Application Output

For one test input, the Streamlit app displayed the following output:

Churn Prediction	No
Churn Probability	34.00%
Explanation	This customer is less likely to leave based on the given details.

11. Project Structure

```
Customer-Churn-Predictor/  
app.py  
requirements.txt  
runtime.txt  
Customer_Churn_Project_Report.pdf  
data/Telco-Customer-Churn.csv  
models/model.pkl  
models/scaler.pkl  
models/columns.pkl  
notebook/churn_analysis.ipynb
```

12. Deployment Details

The project was pushed to GitHub and deployed publicly on Streamlit Cloud. During deployment, dependency compatibility was handled by setting a suitable Python version and package versions. The deployed app is publicly accessible through the live demo link below.

GitHub Repository: <https://github.com/Jaydipnagrani/Customer-Churn-Predictor>

Live Streamlit App: <https://customer-churn-predictor-gw7ugtznrde4xssr5cwq9h.streamlit.app/>

13. Conclusion

This project successfully demonstrates a complete beginner-friendly machine learning workflow and deployment process. A customer churn prediction model was developed, evaluated, saved, converted into a Streamlit web application, pushed to GitHub, and deployed on Streamlit Cloud. The project improved practical understanding of data preprocessing, model evaluation, model serialization, Streamlit UI development, Git/GitHub workflow, dependency management, and cloud deployment.

Final Status: Jupyter Notebook completed, PDF report completed, GitHub repository uploaded, and Streamlit Cloud live deployment completed.